

Figure R — Topology vs physical chords (generator policy + Earth LOS)

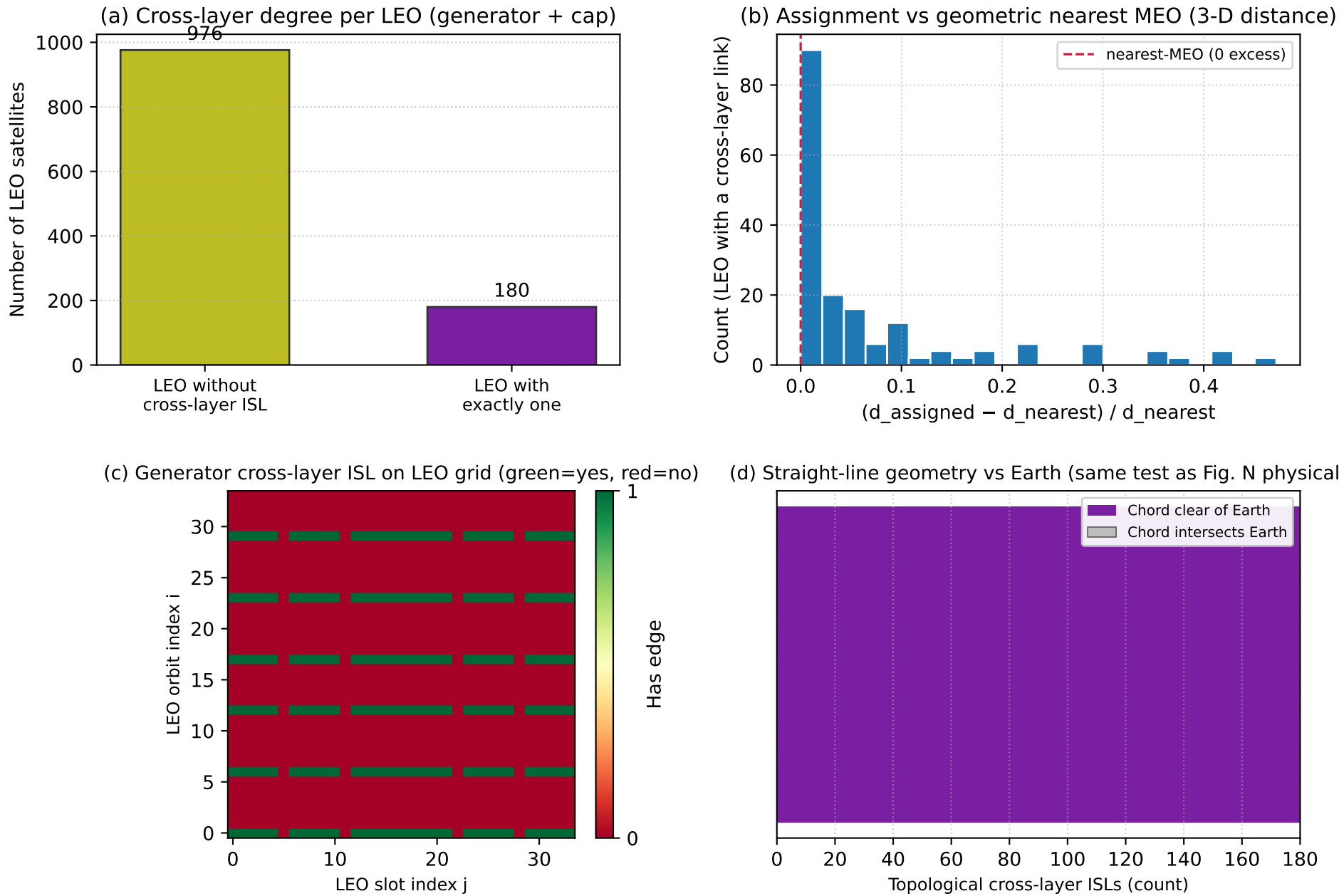


Figure R — Option B: topology (satgen) vs physical chord (Earth sphere)

- Cross-layer rule (not nearest-MEO): $\text{meo_orbit} = \lfloor i \cdot M_{\text{orb}} / L_{\text{orb}} \rfloor$, $\text{meo_slot} = \lfloor j \cdot M_{\text{slot}} / L_{\text{slot}} \rfloor$ then one MEO node; each MEO accepts at most $\text{max_leo_per_meo} = 5$ edges (FIFO). Here 976 LEOs have 0 cross-layer ISL and 180 have 1 (topological total 180; theoretical cap-bound upper bound $\min(\text{LEO}, \text{MEO} \times \text{cap}) = 180$).

- Why some LEOs have no MEO link: the mapped MEO bucket is already full (per-MEO cap), or the generator skips indices past leo_num_sats (not used when the LEO grid is full).

- Earth panel (d): among those topological cross-layer ISLs, 180 chords are strictly outside the Earth sphere; 0 straight chords would intersect the globe in this ECI geometry (Figure N --link-mode physical omits those).

- Figure N still uses matplotlib draw-order for depth; this is not an RF visibility model.